D2F SHIPPING

Functional App Workflow & Operational Design

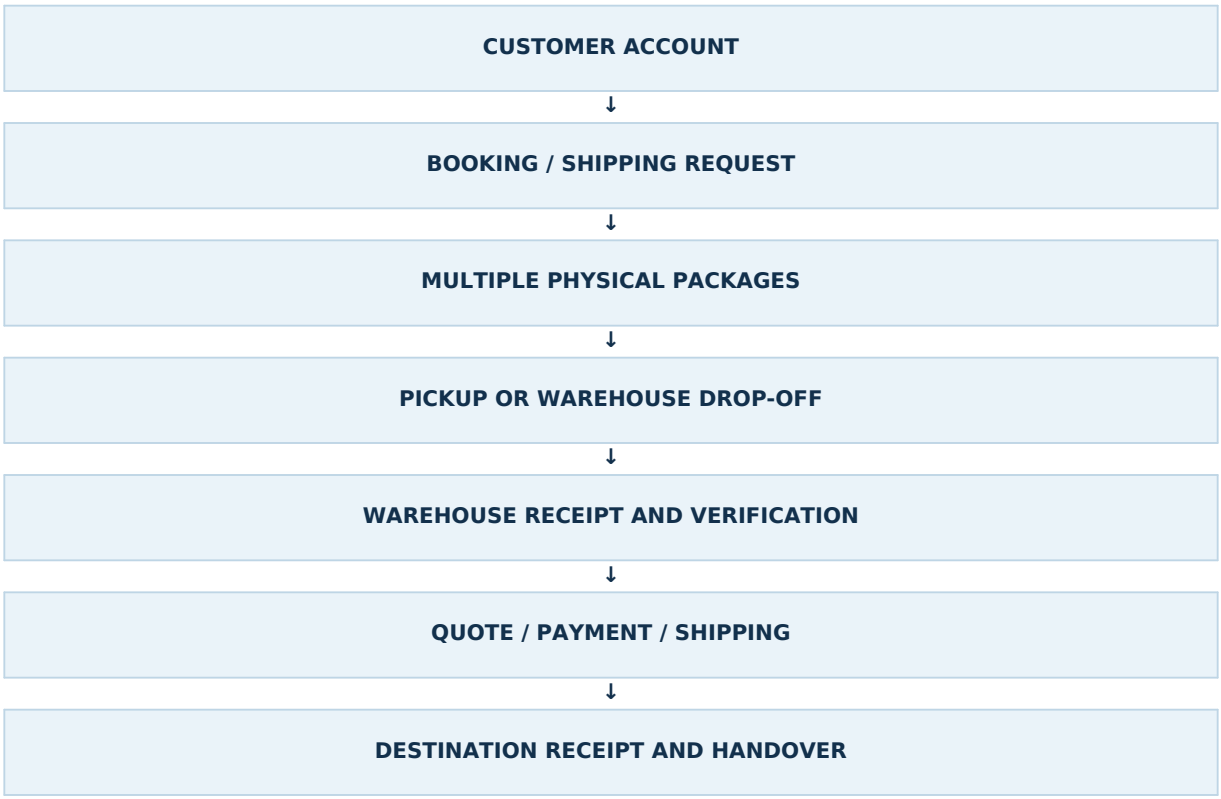
Developer implementation reference | Customer booking, warehouse operations, multi-package shipments and sea container management

Prepared for: Rehan | Operations: Omaha, USA - Lomé, Togo - future West African partners

Purpose: Describe how the D2F application should be structured and how the operational workflow should function. Testing results, implementation feedback and milestone discussions will be communicated separately.

Core architecture

One customer booking can contain multiple physical packages. Each package has its own identity, label and tracking history. A booking may generate one consolidated quotation and invoice. Packages from different bookings may travel together in the same sea container or air shipment.



1. Customer account and shipping request

• Customer registration requires full name, phone number, email address and contact address.

• Each booking records origin, destination, shipping method (sea, air or traveler cargo), recipient name, phone number and destination address.

• Customers may enter addresses manually. Map-based address search/autocomplete is an optional convenience, not a required dependency.

• The customer submits a request before physical receipt or final pricing; request submission does not mean D2F has received the goods.

• The customer must provide truthful descriptions of contents and acknowledge shipping restrictions.

2. Multiple packages under one booking

The customer can add any practical number of packages to one booking: e.g., four cartons, one suitcase and one barrel. A package is a physical handling unit; it can contain multiple declared items.

Booking	Package	Declared contents	Photos
BOOK-001	Box 1	Clothing	Optional
BOOK-001	Box 2	Shoes	Optional
BOOK-001	Suitcase	Personal items	Optional
BOOK-001	Barrel	Household goods	Optional

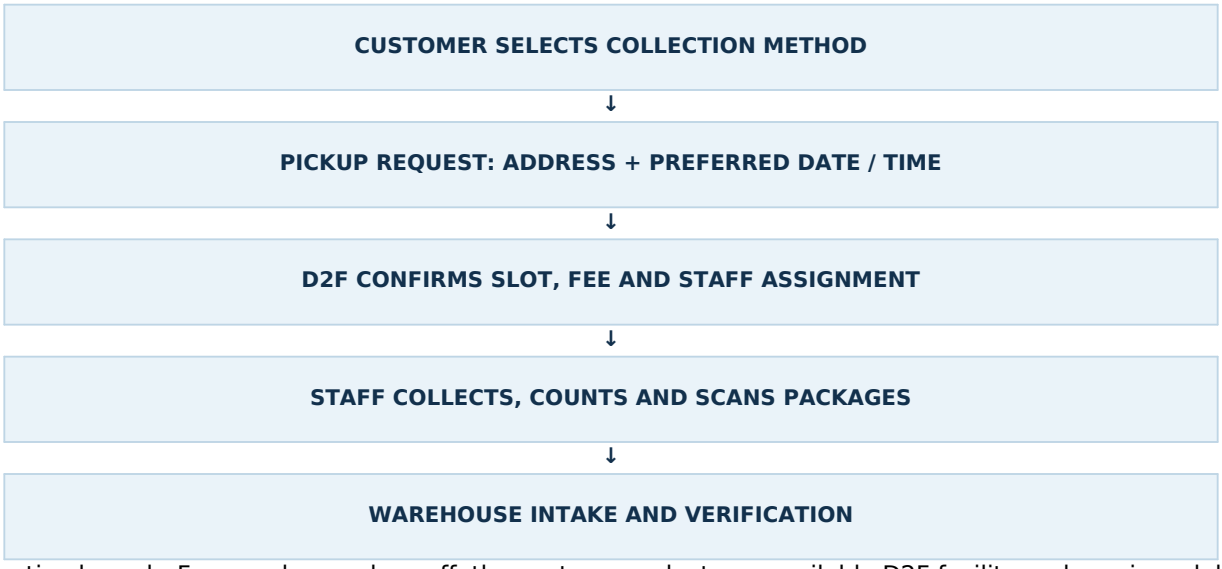
• Each package stores its type, contents description, quantity, optional photos, optional dimensions and mode-dependent weight.

• Each physical package receives a unique package/tracking ID and QR/barcode label when registered for warehouse handling.

• Customer views a consolidated booking with individual package statuses.

• Packages with different recipients or destinations retain separate destination and delivery instructions while remaining linked to the customer order where appropriate.

3. Collection: pickup or drop-off



Alternative branch: For warehouse drop-off, the customer selects an available D2F facility and receives delivery instructions. Warehouse staff record physical receipt only after checking the actual goods.

• Pickup request includes contact number, full pickup address, preferred time window, package count and optional instructions.

• Pickup fee may depend on distance, package count and size. Staff confirm availability and price before collection.

• Support rescheduled, failed, cancelled and partially completed pickups with recorded reasons.

• A request for six packages with only five collected must show the discrepancy; the sixth is not automatically received.

4. Staff and administrator operations

Authorized staff and administrators see an incoming request queue showing customer and recipient details, transport method, declared packages, photos if supplied, addresses and collection choice.

• Review the request and contact the customer if clarification is needed.

• Schedule or confirm pickup, or await warehouse drop-off.

• Record collection and physical warehouse receipt with package count, timestamp, facility and staff identity.

• Inspect contents and condition; document damage, missing or unexpected packages.

• Enter or confirm prices, create a quotation, collect approved payments and prepare labels.

• Assign verified packages to a sea container, air shipment or traveler cargo trip.

• Update operational events only within assigned role and warehouse permissions.

5. Separate status flows

Workflow	Status sequence (illustrative)
Booking	Submitted → Under review → Quoted → Approved → Active → Completed
Pickup	Requested → Scheduled → Collected / Failed / Rescheduled
Warehouse	Declared → Collected (if applicable) → Received → Inspected → Ready for dispatch
Transport	Assigned → Loaded → Departed → In transit → Arrived → Destination received
Handover	Ready for pickup → Released → Delivered

Statuses for booking, pickup, package, transport and delivery are separate records or dimensions. A container arrival at port must not automatically mark an individual package as physically received at the warehouse.

6. Pricing, quotations and invoices

Sea freight: Pricing is primarily based on package size, with optional weight, contents, volume/CBM and staff assessment. The system supports fixed prices, category-based adjustments, custom quotations and authorized manual overrides. Weight is not mandatory for every sea shipment.

Air freight and traveler cargo: Maintain independent weight-, item- and category-based pricing rules; do not apply sea freight assumptions.

Declared value: Record separately from freight price; use for relevant customs, insurance and risk assessment rather than treating it automatically as the shipping charge.



One booking generates a consolidated invoice with separate package lines, pickup fees, adjustments, payments, balance due and payment history. Preserve the agreed pricing snapshot and record who approved changes.

7. Container records and identifiers

The container planning record must be created before the official physical container number is known. The carrier typically supplies a booking reference first. The official container number can be attached later without changing package associations.

Identifier	Role
UUID	Immutable internal database identifier, generated automatically.
D2F sea reference	Unique readable operational record number.
Carrier booking number	Carrier reservation reference, available before physical container assignment.
Official container number	Physical ISO container number, entered when allocated.
Seal number	Recorded when sealed; subsequent changes are logged.

8. Maritime information

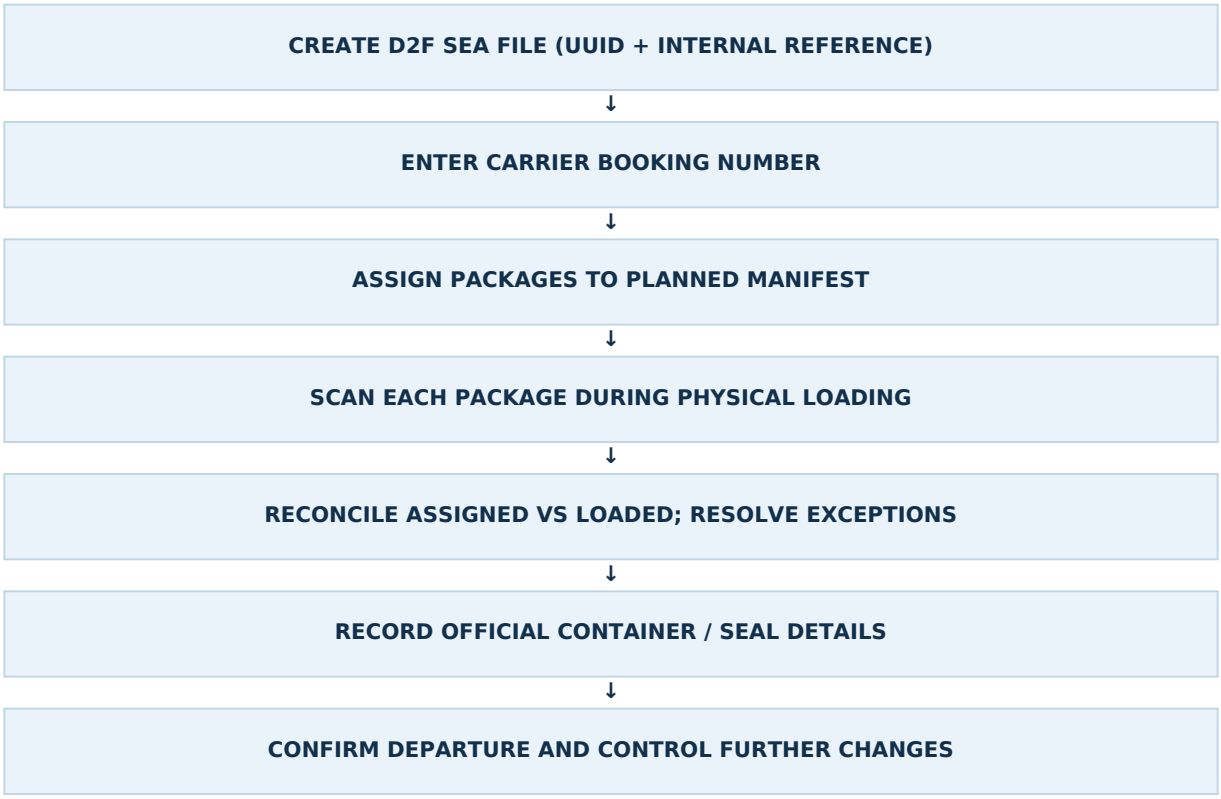
• Carrier, booking reference, container size/type, origin facility, planned receiving facility, discharge port and final destination.

• Vessel and voyage, ETD, ETA, actual departure/arrival, transshipment details and customs milestones.

• Container number and seal number entered or updated as they become available.

• Each change preserves the responsible user and timestamp.

9. Container loading and manifest



• Assigned means planned for the container; Loaded means physically scanned into it.

• The loading workflow prevents duplicate scans and identifies unscanned or incorrectly assigned packages.

• A manifest can be previewed and exported, showing package-level details and totals where available.

• After departure, removals or reassignment require explicit authorized exception handling and an audit trail.

• Package counts are mandatory; weight and CBM totals reflect only data actually recorded and must not be fabricated.

10. Sea tracking, Lomé receipt and onward distribution



The customer tracks each package using its existing D2F tracking number. Container events may inform linked package tracking, but physical receipt, damage, release and delivery require package-level confirmation. Other customers' records and full manifest details remain private.

11. Mixed and future direct routes

Current mixed containers: Every package arrives at D2F Lomé for physical scanning and reconciliation, even if its final recipient is in Benin, Burkina Faso, Ghana or elsewhere. Onward movements are tracked separately.

Future direct-market containers: A shipment may be routed to a partner in Burkina Faso or another market. For landlocked destinations, the ocean discharge port and inland receiving partner are separate locations. The system must support configurable routes rather than requiring Lomé for every future container.

12. Labels, roles and operational safeguards

• Generate one scannable identifier and printable label for each physical package; show package sequence such as “Box 1 of 6” and link to its master booking.

• Support 4 × 6-inch shipping-label output for warehouse printers; mobile Bluetooth printing requires printer-specific integration or a printable PDF fallback.

• Warehouse-specific staff permissions control pickup, loading, receiving, release and delivery; administrators manage cross-warehouse configuration.

• Record recipient name, staff, date/time and appropriate handover confirmation before marking a package delivered.

• Preserve audit logs for status, pricing, package assignment and manifest changes.

• Customers receive timely updates for request review, pickup confirmation, warehouse receipt, dispatch, arrival and delivery.

Operational implementation principle

The customer request, individual physical packages, collection workflow, warehouse receipt, quotation, invoice, transport assignment, container movement and final handover form one connected workflow. The same underlying records must be consistent across the Android customer/staff application and Admin Dashboard.

Document boundary: This is a functional design description for implementation alignment. Testing observations, bug reports, acceptance results, milestone allocation and commercial approvals will be communicated separately after testing.